

Project 4: Highly Available Web Application with ALB

Complete step-by-step implementation guide (AWS Console)

Architecture

Internet -> Application Load Balancer -> Web Server 1 (Availability Zone 1)
-> Web Server 2 (Availability Zone 2)

Step 1: VPC banao

- AWS Console → search “VPC” → open
- “Your VPCs” → “Create VPC”
- “VPC settings” mein “VPC only” select karo
- Name tag: **HA-VPC**
- IPv4 CIDR: **10.0.0.0/16**
- **Create VPC**

Step 2: 2 Public Subnets banao — ALAG Availability Zones mein

- “Subnets” → “Create subnet”
- VPC: **HA-VPC**
- **Subnet 1:** name **PublicSubnet-AZ1**, Availability Zone = pehla wala (e.g. ap-south-1a), CIDR **10.0.1.0/24**
- “Add new subnet” → **Subnet 2:** name **PublicSubnet-AZ2**, Availability Zone = doosra wala, DIFFERENT (e.g. ap-south-1b), CIDR **10.0.2.0/24**
- **Create subnet**

■ **Zaroor check karo:** dono subnets ki AZ column mein alag-alag AZ likhi honi chahiye, warna baad mein ALB banate time problem aayegi.

Step 3: Internet Gateway banao aur attach karo

- “Internet gateways” → “Create internet gateway” → name **HA-IGW** → Create
- Select karo → Actions → “Attach to VPC” → HA-VPC → Attach

Step 4: Route Table banao

- “Route tables” → “Create route table” → name **Public-RT**, VPC HA-VPC → Create
- Select karke “Routes” tab → “Edit routes” → Add route: Destination 0.0.0.0/0, Target = Internet Gateway (HA-IGW) → Save
- “Subnet associations” tab → “Edit subnet associations” → dono subnets tick karo → Save
- Subnets page pe jaake har subnet ke liye: Actions → Edit subnet settings → “Enable auto-assign public IPv4 address” tick karo (dono subnets ke liye alag-alag karna)

Step 5: Security Groups banao

ALB-SG banao:

- EC2 → Security Groups → Create security group
- Name: **ALB-SG**, VPC: HA-VPC
- Inbound rules tab mein (dhyan se — Outbound nahi): Add rule → Type = HTTP, Source = Anywhere-IPv4
- Create security group

WebServer-SG banao:

- Create security group phir se
- Name: **WebServer-SG**, VPC: HA-VPC
- Inbound rules mein 2 rules add karo: Type = HTTP, Source = Custom → ALB-SG select karo | Type = SSH, Source = Anywhere-IPv4
- Create security group

■ **Bahut zaroori:** Rules “Inbound rules” section mein hi add karna, “Outbound rules” mein nahi (dono ek hi page pe hote hain, confuse mat hona). Outbound rules ko bilkul touch mat karna — default “All traffic → 0.0.0.0/0” already sahi hota hai. Rule add karke “Save rules” button zaroor click karna.

Step 6: 2 EC2 Instances launch karo

WebServer-1:

- EC2 → Launch instance
- Name: **WebServer-1**
- AMI: Amazon Linux 2023
- Instance type: **t3.micro** (t2.micro nahi — naye AWS accounts mein free tier eligible nahi hai)
- Key pair: apni key select karo (na ho to pehle bana lo: Key Pairs → Create key pair)
- Network settings → Edit: VPC = HA-VPC, Subnet = PublicSubnet-AZ1, Auto-assign public IP = Enable, Security group = WebServer-SG
- Advanced details → User data mein niche wali script paste karo
- Launch instance

```
#!/bin/bash
dnf install -y httpd
systemctl enable httpd
systemctl start httpd
echo "<h1>Hello from Server 1</h1>" > /var/www/html/index.html
```

WebServer-2: same repeat, bas:

- Subnet: **PublicSubnet-AZ2**
- User data mein “Server 1” ki jagah “Server 2”:

```
#!/bin/bash
dnf install -y httpd
systemctl enable httpd
systemctl start httpd
echo "<h1>Hello from Server 2</h1>" > /var/www/html/index.html
```

Step 7: Target Group banao

- EC2 → Target Groups → Create target group
- Target type: Instances
- Name: **HA-TargetGroup**, Protocol HTTP, Port 80, VPC = HA-VPC → Next
- Dono instances (WebServer-1, WebServer-2) tick karo → “Include as pending below”
- Create target group

Step 8: Application Load Balancer banao

- EC2 → Load Balancers → Create load balancer → Application Load Balancer
- Name: **HA-ALB**, Scheme: Internet-facing, VPC: HA-VPC
- Mappings: dono AZ checkboxes tick karo (ab dono dikhenge kyunki subnets alag AZ mein hain), unke saamne PublicSubnet-AZ1 aur PublicSubnet-AZ2 select karo
- Security groups: ALB-SG (default hata do)
- Listener HTTP:80 → default action = HA-TargetGroup
- Create load balancer

Step 9: Website test karo

- ALB “Active” hone ke baad (~2 min wait), select karke DNS name copy karo
- Browser mein http:// ke saath paste karo
- Refresh karte raho — kabhi “Server 1”, kabhi “Server 2” dikhega

Step 10: High Availability prove karo

- EC2 → WebServer-1 select → Instance state → Stop instance
- Website refresh karo → ab hamesha sirf “Server 2” dikhega
- **Screenshot lo** — ye report/viva ke liye sabse important proof hai

Step 11: Cleanup karo

Isi order mein: Load Balancer delete → Target Group delete → dono EC2 instances terminate → Security Groups delete → Subnets delete → Internet Gateway detach+delete → Route Table delete → VPC delete